

# Tanaka Certificates: Weekly Research Report

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## Abstract

This report presents a Residual Deep ICNN certificate whose local-time sign is safe by construction. For the enlarged-target Ornstein–Uhlenbeck problem, a fixed-feature instance was fitted by linear programming and formally verified on 1,177 cells. A much larger model can be trained with Adam, but exact cell discovery does not scale to that model. Interval bound propagation is therefore the next verification direction. Committor functions do not directly meet the strict certificate conditions, but their transition geometry may help choose training samples.

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## 1 Certificate architecture

Let  $X_t$  satisfy

$$dX_t = f(X_t) dt + \sigma(X_t) dW_t. \quad (1)$$

The certificate  $V$  must be small on the initial set and at least  $\beta$  on the unsafe set and the domain boundary. Before the process reaches the target or the level set  $\{V \geq \beta\}$ , it must also satisfy

$$V \geq 0, \quad \mathcal{L}V \leq -\epsilon. \quad (2)$$

If  $V$  is piecewise smooth, the multidimensional Itô–Tanaka formula adds a local-time term at every interface. The jump in the normal derivative must have the correct sign; checking arbitrary interfaces one by one is expensive.

The Residual Deep ICNN avoids that check by writing

$$\boxed{V(x) = \gamma(S(x) - C(x))}, \quad \gamma > 0. \quad (3)$$

The smooth branch is

$$S(x) = c + p^\top x + \frac{1}{2}x^\top Hx + \frac{1}{2} \sum_{k=1}^m v_k \text{ReLU}(a_k^\top x + b_k)^2. \quad (4)$$

Squared ReLU is continuously differentiable, so  $S$  is  $C^1$  and piecewise quadratic. It has no surface-local-time contribution, but its cellwise Hessian may have either sign. This ordinary curvature is useful because the diffusion part of  $\mathcal{L}V$  depends on second derivatives.

The second branch  $C$  is a fully input-convex ReLU network. With

$$z_1(x) = \text{ReLU}(A_1 x + b_1), \quad (5)$$

$$z_\ell(x) = \text{ReLU}(A_\ell x + W_\ell z_{\ell-1}(x) + b_\ell), \quad W_\ell \geq 0, \quad (6)$$

its output is  $C(x) = a^\top x + b + w^\top z_L(x)$  with  $w \geq 0$ . The direct input weights  $A_\ell$  are unrestricted, while the hidden-to-hidden and output weights are made nonnegative using a softplus parameterization. Since ReLU is convex and nondecreasing, these constraints make  $C$  convex by induction over the layers. It is also continuous and piecewise affine.

Consequently,  $S$  contributes no singular curvature and the kink curvature of  $C$  is positive semidefinite. The singular curvature of  $V$  is therefore

$$D_{\text{sing}}^2 V = -\gamma D_{\text{sing}}^2 C \preceq 0. \quad (7)$$

This gives the safe local-time sign at every ICNN activation interface without enumerating adjacent facets. It does *not* prove the value or ordinary generator inequalities; those still require formal verification.

If  $V \leq \alpha$  on the initial set, these conditions give the reach-avoid bound

$$\mathbb{P}_x(\text{unsafe set or domain exit before target}) \leq \frac{\alpha}{\beta}. \quad (8)$$

## 2 Certificate verified with linear programming

The new verified result uses the harder enlarged-target benchmark

$$dX_t = -X_t dt + 0.5 dW_t \quad (9)$$

on  $[-1, 1.25] \times [-1.25, 0.75]$ . The target is  $[-0.5, 0.5]^2$ ,  $\beta = 2$ , and  $\epsilon = 0.1$ . The initial and unsafe sets are the two small rectangles shown in Figure 1.

A numerical Poisson solution first provides the desired global shape. The ridge locations are then fixed, which makes the unknown piecewise-quadratic coefficients linear. A linear program fits these coefficients to the teacher while enforcing the certificate inequalities. Finally, the exact verifier checks every resulting cell; the sampled diagnostics in the plot are only for visualization.

The verifier returned *verified* with 1,177 cells and no unresolved cells. It proved the thresholds  $\alpha = 1.97$ ,  $\beta = 2$ , and  $\epsilon = 0.1$ , hence the current probability bound is  $\alpha/\beta = 0.985$ . This bound is weak, but the experiment establishes that the architecture, LP fit, and exact verifier work together on the enlarged-target problem.

## 3 Training and verification at larger scale

Linear programming is possible here because the feature geometry is fixed. It is not a practical long-term training method for a deep model whose features must also be learned. The Adam experiment in Figure 2 used 16 smooth hinges and a four-layer width-79 ICNN, for 19,824 trainable parameters. It learned a useful surface, but it did not simultaneously satisfy the initial, unsafe, boundary, and generator conditions. Sampling and counterexample replay improve the fit, yet optimization can move a violation from one condition to another.

For comparison, Figure 3 shows a depth-4, width-80 plain ReLU network with 19,761 parameters. It passed the same dense empirical audit. The signed margins of the two runs are:

| Condition       | Residual Deep ICNN | Plain ReLU |
|-----------------|--------------------|------------|
| Initial         | -0.2704            | +0.0146    |
| Unsafe          | -0.4188            | +2.5601    |
| Domain boundary | -0.4038            | +0.0131    |
| Nonnegative     | +0.0669            | +0.0070    |
| Generator       | -0.3879            | +0.0109    |

This comparison is a negative control, not evidence that plain ReLU is the better certificate. A ReLU network is affine inside each activation cell, so its classical Hessian is zero there and the sampled generator omits diffusion curvature. Its gradient jumps at cell boundaries with no controlled sign, which creates unverified local-time terms. The Residual Deep ICNN has the correct interface sign by construction; the plain ReLU does not.

The immediate verification bottleneck is cell discovery. For 16 smooth ridges followed by four width-79 ReLU layers, a crude region bound is

$$137 \left( 1 + 79 + \binom{79}{2} \right)^4 \approx 1.4 \times 10^{16}. \quad (10)$$

Enumerating activation patterns is therefore infeasible; the present estimate is about  $2 \cdot 10^{15}$  years.

Verifier-guided Poisson-teacher fit: VERIFIED

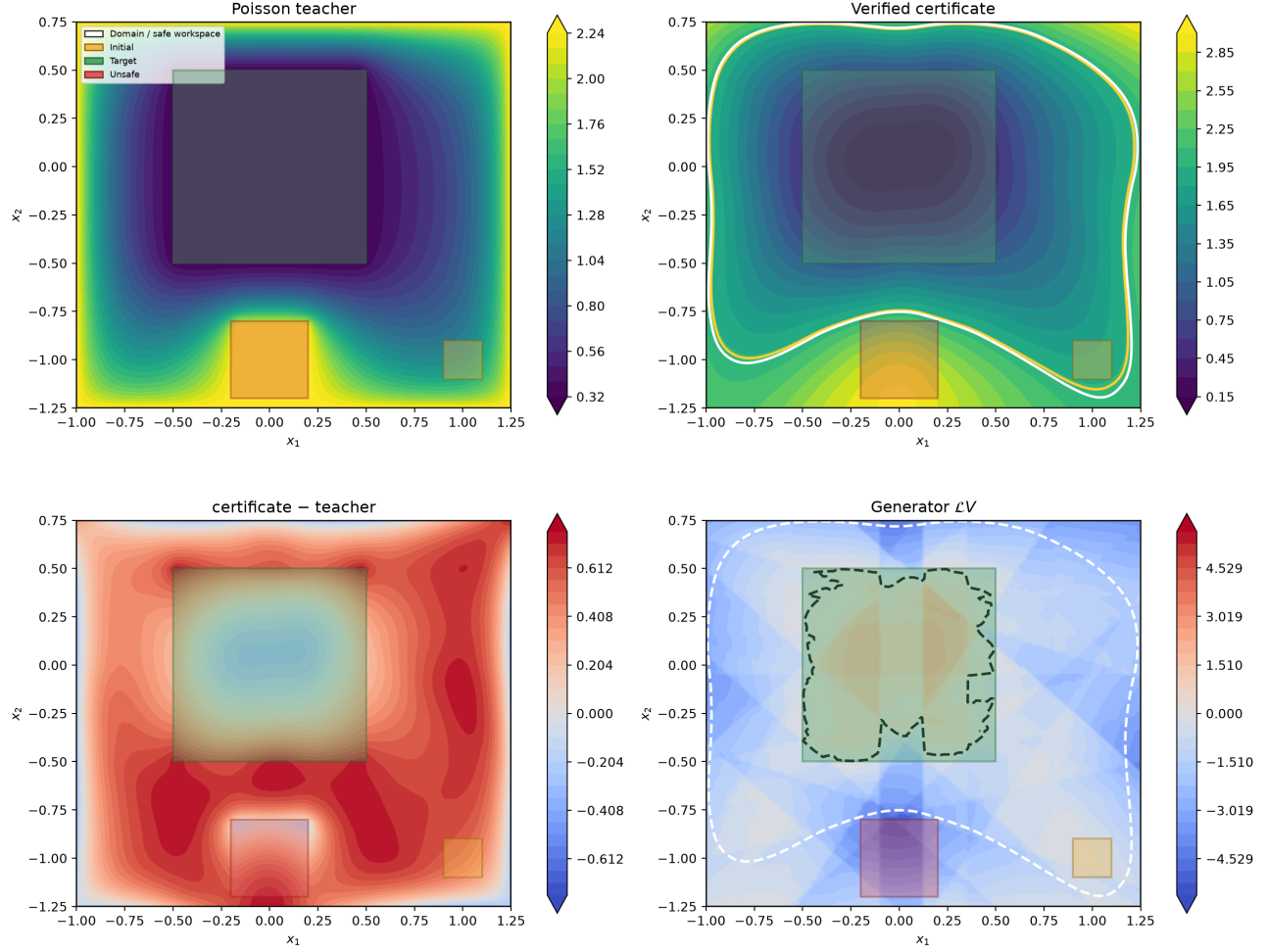


Figure 1: LP-fitted and formally verified certificate. The panels show the Poisson teacher, the certificate, their difference, and  $\mathcal{LV}$ . The white contour is  $V = \beta$ ; the exact result is based on all 1,177 cells, not on the plotted grid.

Adam from scratch:  $\alpha = 1.2$ ,  $\beta = 2$ ,  $\epsilon = 0.1$  — verifier: NOT RUN

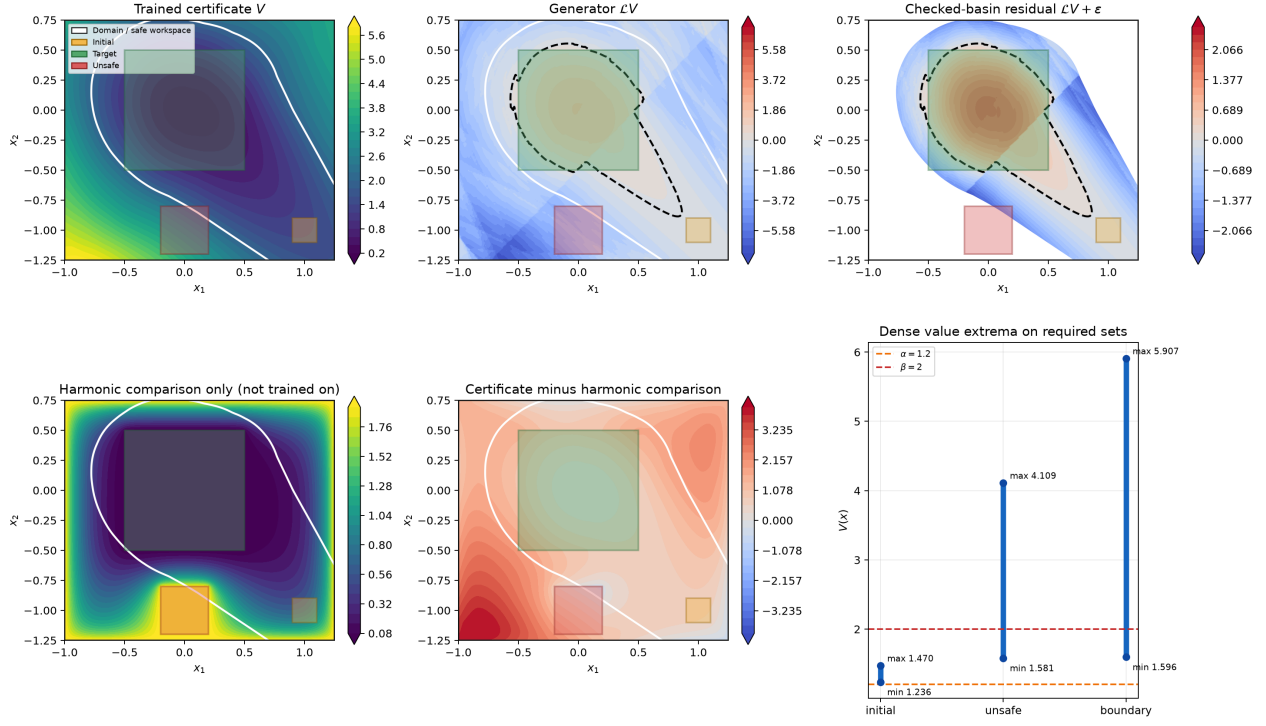


Figure 2: Adam-trained 19,824-parameter Residual Deep ICNN. The local-time sign is structurally safe, but the dense audit still finds value and generator violations. Exact verification was not run because cell discovery is intractable at this size.

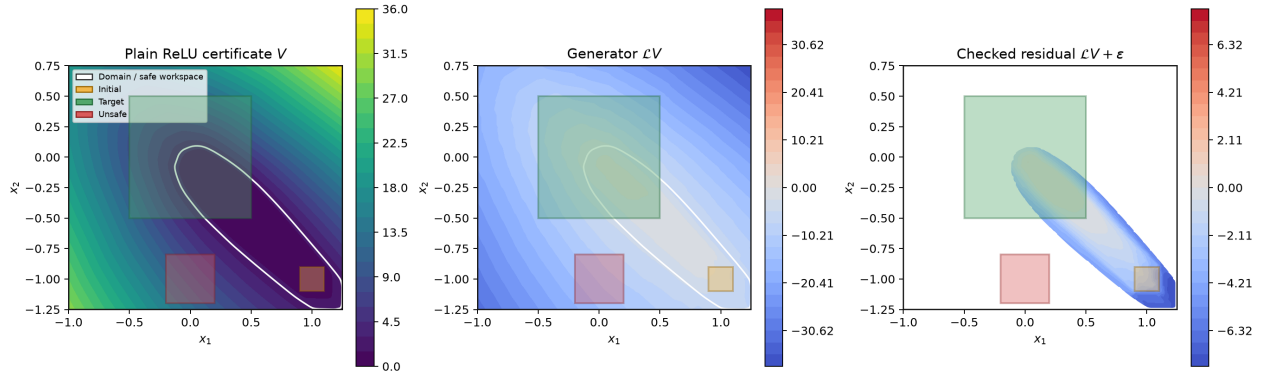


Figure 3: Empirically feasible 19,761-parameter plain ReLU network. Its pointwise plots do not include activation-interface local time, so this is not a valid stochastic certificate despite passing the dense audit.

The next approach is interval bound propagation (IBP) with adaptive splitting. For a box  $B$ , verification only needs to prove

$$\underline{V}(B) > \beta \quad \text{or} \quad \overline{\mathcal{L}V}(B) \leq -\epsilon. \quad (11)$$

Ambiguous boxes are split until one inequality holds. ReLU bounds are simple and cheap, whereas smooth activations generally need more expensive nonlinear relaxations. The local-time sign remains guaranteed by the architecture and does not enter this branch-and-bound search.

## 4 Committor functions as a sampling guide

For two disjoint sets  $A$  and  $B$ , the committor is

$$q(x) = \mathbb{P}_x(\tau_B < \tau_A). \quad (12)$$

In this project, let  $A$  be the target and let  $B$  be failure: the unsafe set or domain exit. Then  $q(x)$  is simply the probability of failure before success from state  $x$ . It equals 0 on the target, 1 on the failure boundary, and satisfies  $\mathcal{L}q = 0$  between them. Values near  $q = 1/2$  identify states for which both outcomes are plausible, so these states contain more useful transition information than points deep inside either region.

For reversible overdamped Langevin dynamics, write the inverse temperature as  $\kappa$  to distinguish it from the certificate threshold  $\beta$ :

$$dX_t = -\nabla U(X_t) dt + \sqrt{2\kappa^{-1}} dW_t, \quad \mathcal{L} = -\nabla U \cdot \nabla + \kappa^{-1} \Delta. \quad (13)$$

Let  $D = \Omega \setminus (A \cup B)$ . The committor is equivalently the minimizer of the weighted Dirichlet energy

$$q = \underset{u \in H^1(D)}{\operatorname{argmin}} \underbrace{\int_D \|\nabla u(x)\|^2 e^{-\kappa U(x)} dx}_{\mathcal{E}(u)}, \quad u|_{\partial A} = 0, \quad u|_{\partial B} = 1, \quad (14)$$

with the appropriate condition on the outer boundary of  $\Omega$ . This optimization has a probabilistic interpretation: among functions separating  $A$  and  $B$ , the committor has minimum energy under the equilibrium weight  $e^{-\kappa U}$ .

The connection to the backward equation follows from the first variation. For any perturbation  $\eta$  that is zero on  $\partial A \cup \partial B$ ,

$$0 = \left. \frac{d}{ds} \mathcal{E}(q + s\eta) \right|_{s=0} \quad (15)$$

$$= 2 \int_D \nabla q \cdot \nabla \eta e^{-\kappa U} dx = -2 \int_D \eta \nabla \cdot (e^{-\kappa U} \nabla q) dx. \quad (16)$$

Hence  $\nabla \cdot (e^{-\kappa U} \nabla q) = 0$ , which is equivalent to  $\mathcal{L}q = 0$ . This also shows why the variational loss is attractive for neural networks: it needs only first derivatives of  $q$ , whereas a direct PDE residual  $\|\mathcal{L}q\|^2$  needs second derivatives.

With a neural approximation  $q_\theta$  and interior sampling density  $r$ , a Monte Carlo training objective has the form

$$J(\theta) = \mathbb{E}_{X \sim r} \left[ \frac{\|\nabla q_\theta(X)\|^2 e^{-\kappa U(X)}}{r(X)} \right] \quad (17)$$

$$+ \lambda_A \mathbb{E}_{X \sim \partial A} [q_\theta(X)^2] + \lambda_B \mathbb{E}_{X \sim \partial B} [(q_\theta(X) - 1)^2]. \quad (18)$$

The last two terms enforce the boundary values. The interior integrand  $\|\nabla q_\theta\|^2 e^{-\kappa U}$  is also an error indicator: it is large where the current model places transition energy. DASTR therefore fits a generative model to the normalized density

$$p_{U, q_\theta}(x) \propto \|\nabla q_\theta(x)\|^2 e^{-\kappa U(x)} \quad (19)$$

and uses its samples in the next optimization round. This exact variational form relies on reversible Langevin dynamics; a general non-reversible SDE requires a different weak formulation or a residual loss.

Figure 4 gives a concrete example. The stochastic dynamics spend most of their time near the metastable sets  $A$  and  $B$ , while the committor changes between 0 and 1 across the region separating them. The neural approximation reproduces this global transition geometry. The important difficulty is visible in the first panel: ordinary trajectory data are concentrated near  $A$  and  $B$ , not where the committor changes most.

DASTR addresses this data problem with the feedback loop in Figure 5. A provisional committor determines a sampling density that emphasizes the transition region. New samples from that density are used to retrain the

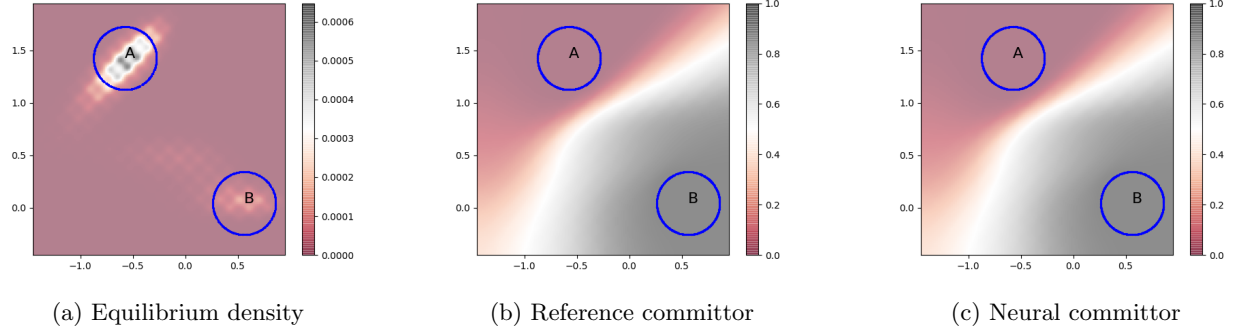


Figure 4: Rugged Müller example at temperature  $T = 22$ . The density is concentrated in the metastable regions, whereas the committor describes the transition from  $A$  to  $B$ . Reproduced from Figure 6 of [Khoo, Lu, and Ying](#).

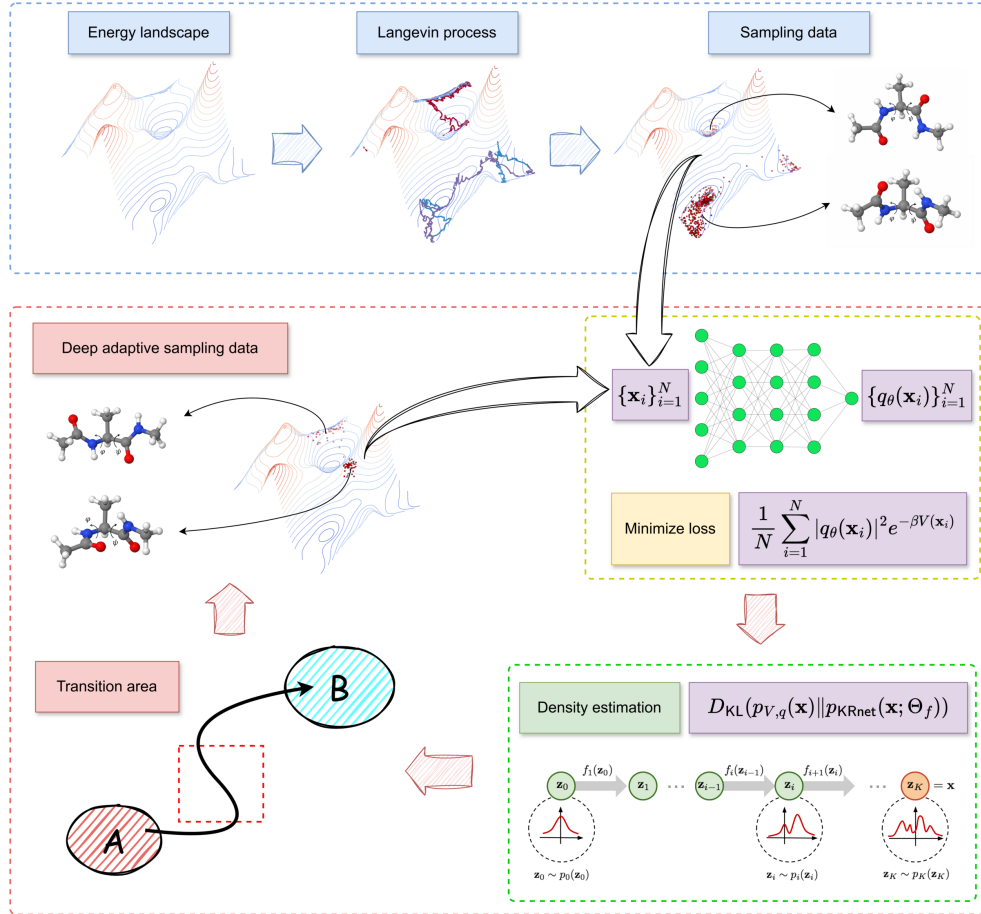


Figure 5: The DASTR feedback loop: ordinary dynamics undersample rare transitions, so a generative density model produces additional transition points for the next committor update. Reproduced from Figure 1 of [Wang et al.](#)

committor, and the process is repeated. This is closely related to counterexample replay in certificate training: the current model decides where the next training points should be placed.

The representation used by the sampler also matters. In Figure 6, direct sampling in the full molecular coordinate space breaks correlations between atoms and produces invalid configurations. Sampling in a lower-dimensional space of latent collective variables preserves the relevant molecular geometry. The same lesson applies to certificate training: adaptive samples must both target difficult regions and remain valid states of the original problem.

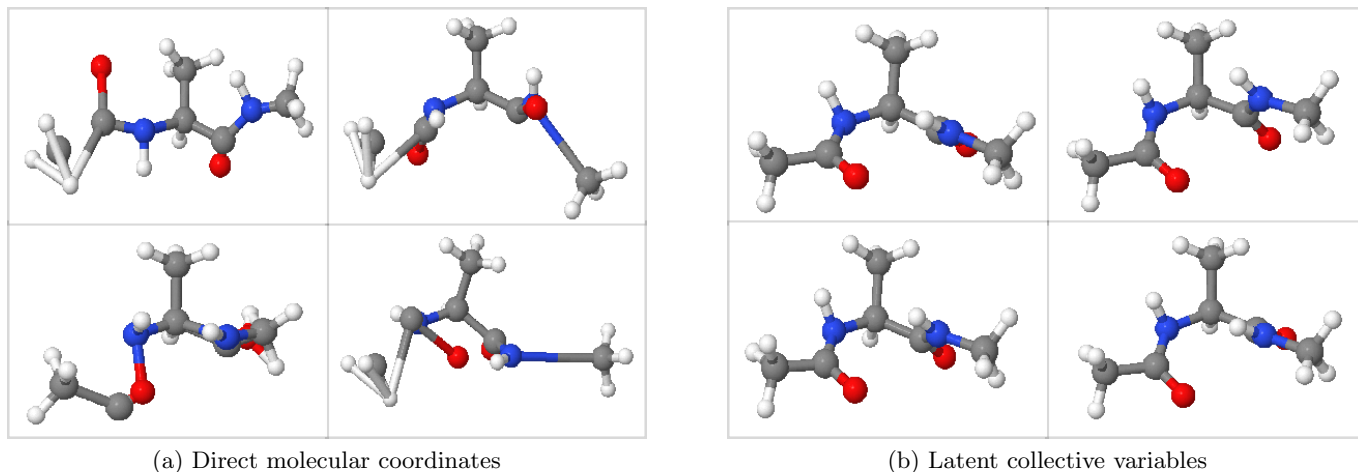


Figure 6: Samples generated in two DASTR representations. Direct-coordinate sampling gives mostly invalid structures; latent-coordinate sampling gives physically consistent structures. Reproduced from Figure 2 of [Wang et al.](#)

The same principle can support certificate training. A learned committor can identify transition regions, and training can place more points there together with known verifier counterexamples. However, the committor itself is not a certificate: it satisfies  $\mathcal{L}q = 0$ , while the certificate requires the strict inequality  $\mathcal{L}V \leq -\epsilon$ . It should therefore be used only as a teacher or sampling guide. The final guarantee must still come from a construction-safe model and a formal verifier.

## 5 Summary and takeaways

1. The Residual Deep ICNN architecture is still promising, since:
  - (a) We don't have to check the local-time sign condition on every facet
  - (b) I managed to formally verify it by optimizing with **Linear Programming** (infeasible in the long run.)
  - (c) I managed to optimize it with **Adam**, but required 100x parameters and couldn't verify it formally (it looks ) (feasible in the long run.)
2. I should have listened to Greg and used IBP instead of cell discovery for verification. Cell discovery would take  $2 \cdot 10^{15}$  years.
3. IBP through ReLUs is still potentially faster than through smooth activations
4. Commitor functions are not directly useful here, but they can help with training hard certificates by guiding sampling